



EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examinations 2021
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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BIOLOGY

9744/04

Paper 4 Practical

24 August 2021

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use stapler, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided in the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
Total	55

This document consists of **17** printed pages and **3** blank pages.

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Answer **all** questions.

Plan the use of the time given to ensure that you finish the whole of Question 1 and Question 2.

- 1** Yeast cells contain respiratory enzymes that catalyses the oxidation of sugars to release energy. During these reactions, carbon dioxide is released. Some of this carbon dioxide dissolves in water forming carbonic acid.

The progress of these enzyme-catalysed reactions can be followed by measuring the time taken for an indicator to change colour as carbonic acid is formed. This gives an indication of the rate of respiration in yeast cells.

You will investigate the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour.

You are provided with the materials shown in Table 1.1.

Table 1.1

labelled	contents	hazard	volume/cm ³
Y	10% yeast suspension	none	100
H	hydrogencarbonate indicator solution	none	50
C	carbonic acid solution	eye irritant	5
W	distilled water	none	100

You are also provided with:

- a beaker, labelled **hot water**, in which to collect hot water at temperature of 50 °C to 55 °C
- a beaker, labelled **tap water**, in which to collect tap water

If any solution comes into contact with your skin, wash it off immediately under cold water. It is recommended that you wear suitable eye protection. Care should be taken when using hot water.

The hydrogencarbonate indicator solution, **H**, will change colour from red to yellow as carbon dioxide passes through it.

To help you identify the end-point, carry out step 1 to step 3.

- 1** Put 2 cm³ of **H** into a small test-tube.
- 2** Add 1 drop of **C** into this test-tube and mix.
- 3** The colour of the indicator solution will change from red to yellow. This is the end-point.

You will need to use this test-tube for colour comparison.

- (a)** You will use proportional dilution to make different concentrations of yeast suspension, Y .

You will need to prepare 25 cm³ of each concentration of yeast suspension.

- (i) State the concentrations of yeast suspension that you will use in your investigation.

..... [1]

- (ii) Complete Table 1.2 to show how you will prepare the concentrations of yeast suspension you have listed in (a)(i).

Table 1.2

concentration of yeast suspension /%	volume of Y /cm ³	volume of W /cm ³
10.0	25.0	0.0

[2]

- 4** Use a glass rod to stir the yeast suspension, **Y**, to suspend the yeast cells. When using a syringe to transfer **Y**, remove **Y** from below any froth on the surface.

- 5** Prepare the concentrations of yeast suspension as stated in **(a)(i)** and shown in Table 1.2.

A student carried out some preliminary experiments to find the optimum conditions for investigating the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour.

The student found that:

- the yeast suspension should **not** be mixed directly with hydrogencarbonate indicator, as the colour change was difficult to observe
- the optimal temperature for yeast enzyme activity is 40 °C
- the best volume of yeast suspension to use is 20 cm³
- the best volume of hydrogencarbonate indicator to use is 2 cm³.

- the different concentrations of yeast suspension from step **5**
- the test-tube showing the end-point colour from step **3**
- hydrogencarbonate indicator solution.

Your planned method should:

- have a clear and helpful structure so that the method described could be repeated by anyone reading it
- include helpful details to ensure that results are as accurate and repeatable as possible
- use the correct technical and scientific terms
- only make use of the apparatus and materials provided.

[illegible]

.....

.....

.....

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.....

..... [6]

Before proceeding further, use the beaker labelled **hot water** to collect approximately 400 cm³ of hot water from where it is provided in the laboratory.

The beaker labelled **tap water** can be filled as necessary from the tap.

Proceed as follows.

- 6** Carry out your method described in **(a)(iii)** to investigate the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour.

Record your results in **(a)(iv)**.

If no colour change occurs with hydrogencarbonate indicator solution after three minutes, record the result as 'more than 180'.

- (iv)** Use the space provided to record your results from step **6** in a suitable format.

- (v) From your results, state a valid conclusion about the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour.

.....
..... [1]

- (vi) Identify one significant source of error in this investigation. Give a reason for your answer.

source of error

reason

..... [2]

- (b) A student carried out an investigation into the metabolic activity of enzymes in 5 different species of yeast, **P** to **T**.

The student immobilised the yeast cells in alginate beads, which were then dropped into a solution of the substrate at the same temperature.

As carbon dioxide collected in the beads, they rose to the surface of the solution of the substrate as shown in Fig. 1.1.

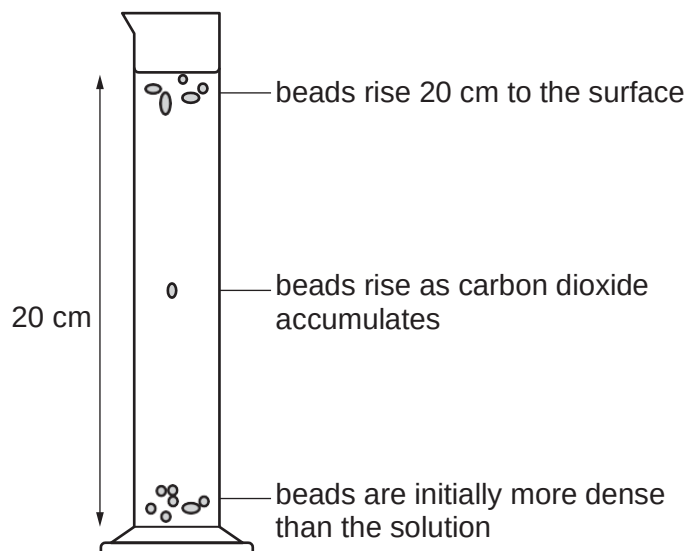


Fig. 1.1

The student measured the time taken for 5 beads to rise to the surface, a distance of 20 cm.

The processed results for each species are shown in Table 1.3.

Table 1.3

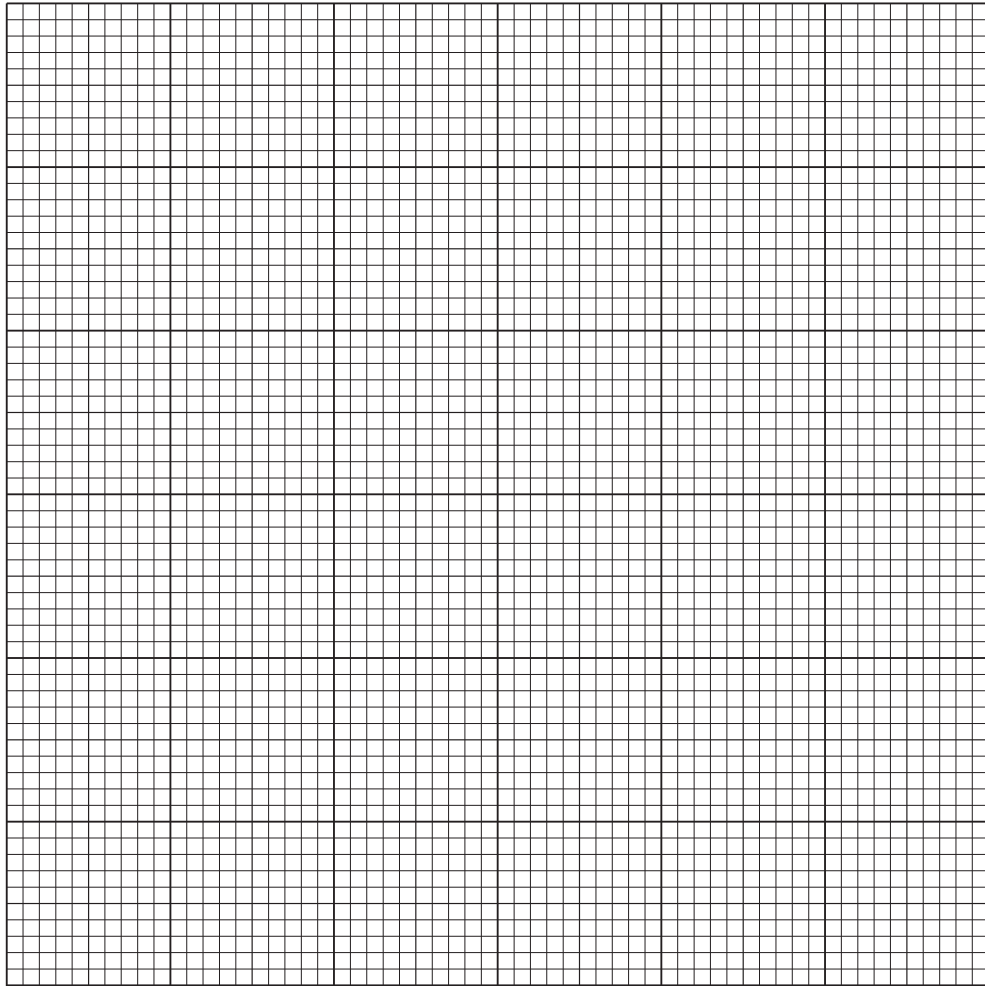
species	mean time to rise / s
P	72
Q	19
R	11
S	36
T	108

- (i) Identify the species with the most active metabolic enzymes [1]

For this species, calculate the rate at which these beads rise 20 cm.

rate = [2]

- (ii) Use the grid to display the results shown in Table 1.3 in an appropriate form.



[4]

- (iii) Suggest why the results differ between species **R** and **T**.

.....

.....

.....

.....

.....

..... [3]

- (c) Methylene blue stains dead cells, but is decolourised by living cells so they will appear white or clear.

You are required to observe the colour of methylene blue when added to three different yeast suspensions, **A1**, **A2** and **A3**.

It is known that each yeast suspension may be one of the following:

- boiled yeast,
- yeast in a high concentration of sodium chloride solution,
- yeast in a glucose solution.

Using this information, it is possible to deduce the identity of each yeast suspension.

Proceed as follows.

- 1 Label three microscope slides **A1**, **A2** and **A3**.
 - 2 Place **one drop** of **A1** onto slide **A1** and add **one drop** of methylene blue.
 - 3 Use a glass rod to mix carefully.
 - 4 Leave for five minutes.
 - 5 Repeat steps 2 to 4 with samples **A2** and **A3**.
 - 6 Add a coverslip to each slide.
 - 7 Use the paper towel to dry off any liquid around the coverslip.
 - 8 View the slides using the microscope.
- (i) Deduce the identity of each yeast suspension by filling in the table below with **A1**, **A2** or **A3**.

description	yeast suspension
boiled yeast	
yeast in a high concentration of sodium chloride solution	
yeast in a glucose solution	

[2]

- (ii) Explain the difference in appearance of the yeast cells in **A1** and **A2**.

.....

.....

.....

..... [2]

[Total: 29]

Question 2 starts on page 12

2 L1 is a slide of a stained transverse section through a plant stem.

You are not expected to be familiar with this specimen.

(a) Select a field of view so that you can observe a vascular bundle on L1.

Use a sharp pencil for drawing.

You are expected to draw the correct shape and proportions of the different tissues.

(i) Draw a large plan diagram of a vascular bundle on L1.

Use **one** ruled label line and label to identify the xylem.

[5]

- (ii) Observe the sclerenchyma cells of the stem section on **L1**. These cells are stained red and are located towards the periphery of the stem section, as indicated in Fig. 2.1.

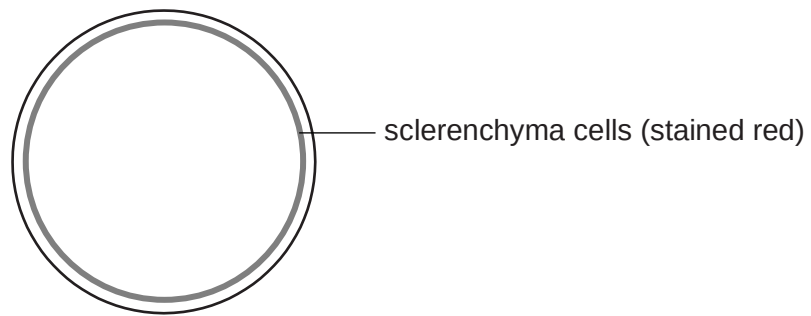


Fig. 2.1

Select a **group** of **four** adjacent sclerenchyma cells.

Each cell must touch at least two other cells.

- Make a large drawing of this group of **four** adjacent cells.
- Use **one** ruled label line and label to identify the cell wall of **one** cell.

- (b) Fig. 2.2 is a photomicrograph of a stained transverse section through a stem of a different type of plant.

You are not expected to be familiar with this specimen.

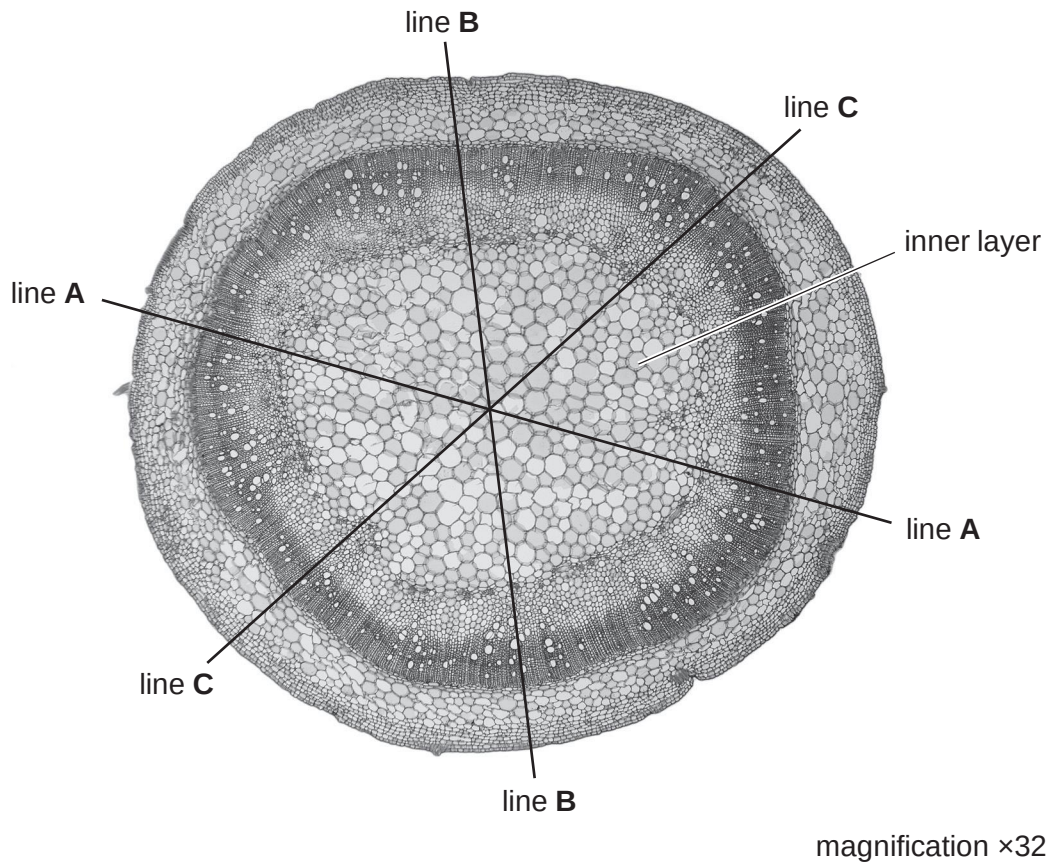


Fig. 2.2

- (i) Use line **A**, line **B** and line **C** to determine:
- the mean actual diameter of the whole stem section
 - the mean actual diameter of the inner layer (pith).

Show all the steps in your working.

mean actual diameter of whole stem section

mean actual diameter of inner layer

- (ii) Use your answers to **(b)(i)** to determine the simplest whole number ratio of the mean actual diameter of the whole stem section to the mean actual diameter of the inner layer.

simplest whole number ratio = [1]

- (c) Fig. 2.3 is a photomicrograph of the same stem section that is in Fig. 2.2.



Fig. 2.3

Identify the observable differences between the stem section in Fig. 2.3 and the stem section on **L1**.

Record **two** observable differences in Table 2.1.

Table 2.1

feature	Fig. 2.3	L1

[2]

- (d) In plants the growth regulator, auxin, is synthesised in the stem tip and moves away from the tip. The movement of auxin through plant tissues was investigated using bean seedlings as shown in Fig. 2.4.

The following procedure was used.

- Stems were cut into 60 mm lengths.
- Agar blocks containing radioactive auxin were placed on the apical surfaces of two groups of stem lengths.
- The basal ends of the stem lengths were placed on agar blocks without any auxin to provide support.

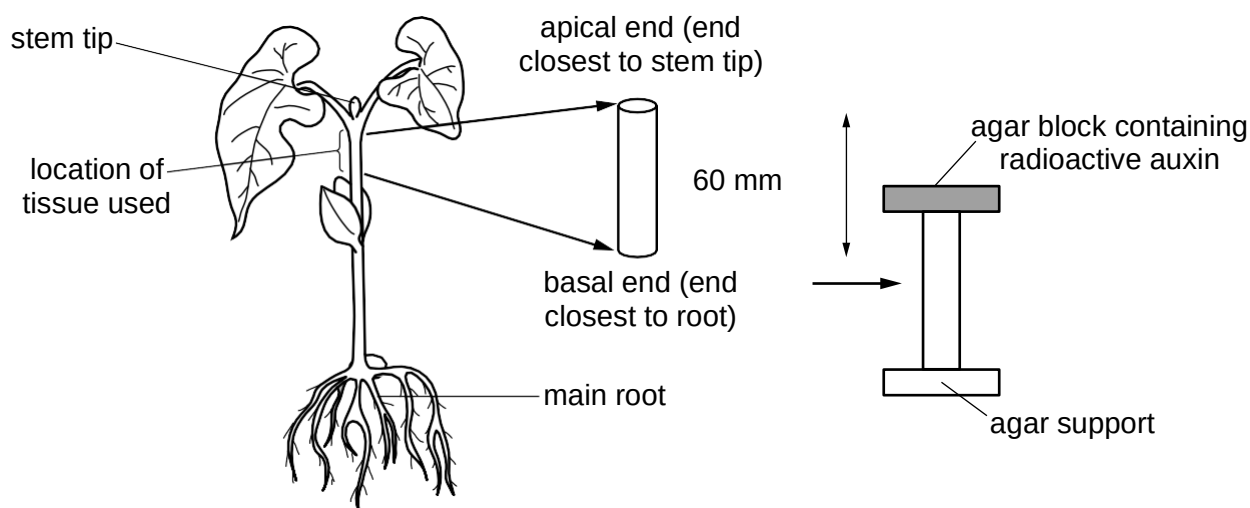


Fig. 2.4

- After 10 minutes the agar blocks at the apical ends were removed.
- One group of stem lengths was placed in air and the other group in an atmosphere of nitrogen.
- Both groups were left in light for 30 minutes after removing the agar blocks.
- The position of the radioactivity was located.

Fig. 2.5 shows the results of the investigation.

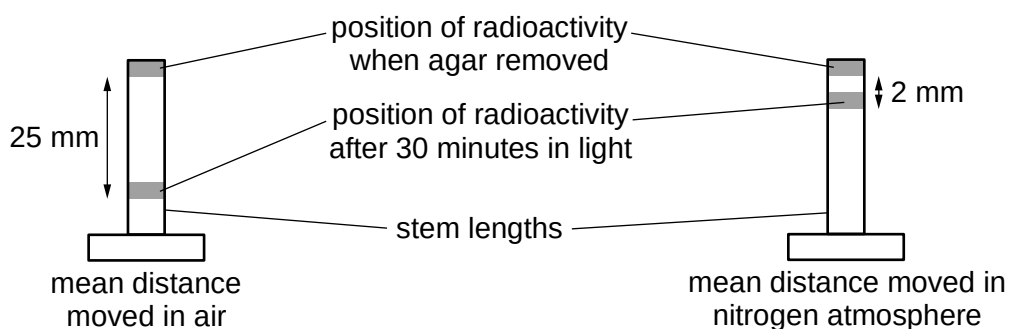


Fig. 2.5

- (i) Identify one variable that should be standardised during this investigation.

.....[1]

(ii) Suggest **one** conclusion that can be made from these results.

.....
 [1]

(iii) Calculate the rate in mm h^{-1} of movement of auxin for the setup placed in air.

[1]

A similar investigation was carried out to test the hypothesis:

The rate of movement of auxin will be faster in plants grown in the light than plants grown in the dark.

Table 2.2 shows the results of this investigation.

Table 2.2

plants grown in light										plants grown in the dark									
sample number										sample number									
1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8	9	10
rate of movement / mm h^{-1}																			
56	61	66	52	50	68	76	51	55	64	45	52	42	35	55	38	32	37	45	51
mean $\pm$ standard deviation (s) = 59.9 ± 8.5										mean $\pm$ standard deviation (s) = 43.2 ± 7.7									

(iv) Describe how the following evidence from Table 2.2 supports the hypothesis.

mean

 standard deviation
 [2]

- (v) A t -test was carried out to see if the difference in the rates of movement of auxin in plants grown in the light and plants grown in the dark was significant.

Suggest a null hypothesis for this statistical test.

.....
..... [1]

- (vi) Explain how the student should use the value for t to find out if the difference in the rates of movement of auxin is significant.

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.....
..... [3]

[Total: 26]

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